

① Dominant kernel: 2D Convolution

Dimensions: Input Image: $N=28$ 28×28

Weight Filter: $K=3$ 3×3

Output Image: $M=26$ 26×26

Data Type: INT8 input

INT8 weight

INT32 output

② Total ops: $2 \times M^2 \times K^2$

$$= 2 \times 26^2 \times 3^2$$

$$= \boxed{12,168 \text{ Ops}}$$

③ Lower bound: Read bytes = $2 \times M^2 \times K^2 = 2 \times 26^2 \times 3^2 = 12,168$

$$\text{Write bytes} = 4 \times M^2 = 4 \times 26^2 = 2706$$

$$\text{Total bytes} = 12,168 + 2706 = 14,872 \text{ bytes}$$

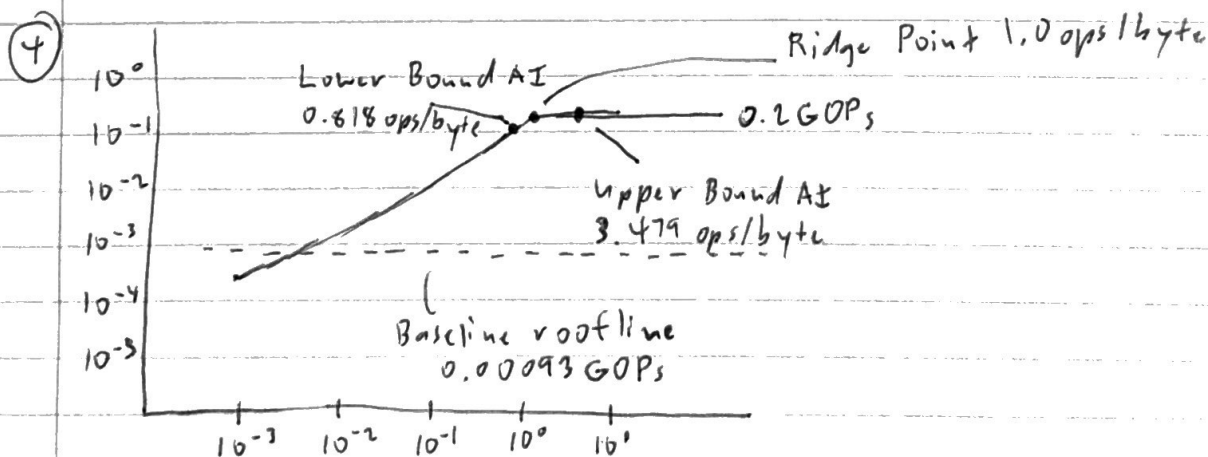
$$\text{Lower bound AI} = 12,168 \text{ ops} / 14,872 = \boxed{0.818 \text{ Ops/byte}}$$

Upper bound: Read bytes = $K^2 + N^2 = 3^2 + 28^2 = 793$

$$\text{Write bytes} = 4 \times M^2 = 4 \times 26^2 = 2706$$

$$\text{Total bytes} = 793 + 2706 = 3497 \text{ bytes}$$

$$\text{Upper bound AI} = 12,168 / 3497 = \boxed{3.479 \text{ Ops/byte}}$$



⑤ The design is currently limited by the hardware interface bandwidth, making it memory bound.

• The highest-leverage change to improve performance would be implementing a local SRAM weight buffer.